APPLIED PHYSICS CORE COURSES & ELECTIVES | GENERAL M.S. DEGREE

(10) Courses: (4) Core Courses [at least (4) from the first six listed] + (6) Electives [including at least (3) 700- or 800-level]

CORE COURSES

615.641
Mathematical Methods for Physics & Engineering
Every Term • Asynchronous

615.653
Classical Mechanics
Every Term • Asynchronous

615.671
Principles of Optics
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

615.651
Statistical Mechanics and Thermodynamics
Every Term • Asynchronous

615.665
Modern Physics
SP / SU • Asynchronous

675.613
Bold Science
Every Term • Asynchronous

615.661
Introduction to Planetary Science
SP / SU • Asynchronous

615.662
Introduction to Astrophysics
FA • Asynchronous

615.663
Physics of Lunar Exploration
In Development • Async.

615.664
Introduction to Exoplanets
In Development • Async.

615.666
Earth Systems & AI
In Development • Async.

615.682
Materials Characterization
In Development • Async.

615.683
Materials in Extreme Environments
In Development • Async.

615.685
Quantum Measurement and Sensing
In Development • Async.

615.715
Applied Atomic and Molecular Physics
Spring • Asynchronous

615.744
Physics of Space Systems I
SU / FA • Asynchronous

615.745
Physics of Space Systems II
SP • Asynchronous

615.747
Sensors and Sensor Systems
FA / SP • Asynchronous

615.748
Introduction to Relativity
SP • Asynchronous

615.749
Applied Nuclear Physics
In Development • Async.

615.751
Modern Optics
Fall • Asynchronous

615.753
Plasma Physics
SU / FA • Asynchronous

615.755
Space Physics
FA / SP • Asynchronous

615.757
Solid State Physics
FA • Asynchronous

615.760
Physics of Semiconductor Devices
SP • Asynchronous

615.761
Introduction to Oceanography
SP • Asynchronous

615.762
Applied Computational Electromagnetics
SP • Synchronous Online

615.763
Infrared Sensors: Physics and Applications
SP • Asynchronous

615.765
Chaos and Its Applications
SU / FA • Asynchronous

615.769
Physics of Remote Sensing
SP • Asynchronous

615.772
Cosmology
SU • Asynchronous

615.775
Physics of Climate
SP • Asynchronous

615.776
Quantum Algorithms: Near Term Applications
SU / SP • Asynchronous

615.778
Optical System Design and Modelling
Every Term • Asynchronous

615.780
Optical Detectors and Applications
FA • Asynchronous

615.781
Quantum Information Processing
SP • Asynchronous

716.782
Optics and MATLAB
SP • Asynchronous

615.783
Experimental Quantum Computing
In Development • Async.

615.784
Superconducting Devices: Physics and Applications
FA / SU • Asynchronous

615.786
Introduction to Open Quantum Systems
In Development • Async.

615.800
Applied Physics Project
Every Term • Asynchronous

615.802
Directed Studies in Applied Physics
Every Term • Asynchronous

Test your degree path with the Applied Physics Course Mapping Tool